Practice paper 1 – Exercise 1

Integration and L^p spaces | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. For $n \geq 1$, let $f_n(x) = x/n$ on $(0,1)$, equipped with Lebesgue measure.

(a) [2 points] Find the pointwise limit of f_n and compute $\int_0^1 f_n(x)\,dx$.

(b) [3 points] Compute $\|f_n\|_{L^1}$ and $\|f_n\|_{L^2}$. Does f_n converge to zero in each of these spaces? *Recall:* $\|h\|_{L^1} = \int_0^1 |h|\,dx$ and $\|h\|_{L^2} = (\int_0^1 |h|^2\,dx)^{1/2}$.

(c) [2 points] Show that $|f_n(x)| \leq 1$ on $(0,1)$. Verify the hypotheses of the dominated convergence theorem and use it to justify interchanging limit and integral.

Practice paper 1 – Solution 1

Detailed worked solution

(a) Pointwise limit and integral. Fix $x \in (0,1)$. Since x is fixed and $n \to \infty$, $x/n \to 0$. Thus the pointwise limit is $f(x) = 0$.

$$\int_0^1 f_n(x)\,dx = \frac{1}{n}\int_0^1 x\,dx = \frac{1}{n}\left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2n} \longrightarrow 0.$$

The integral of the limit is $\int_0^1 0\,dx = 0$, so the two results agree.

(b) Norms and convergence. As $f_n \geq 0$, its L^1 norm equals its integral:

$$\|f_n\|_{L^1} = \frac{1}{2n}.$$

For the L^2 norm, square the function, integrate, and finally take the square root:

$$\|f_n\|_{L^2}^2 = \frac{1}{n^2}\int_0^1 x^2\,dx = \frac{1}{3n^2}, \qquad \|f_n\|_{L^2} = \frac{1}{\sqrt{3}\,n}.$$

Both norms tend to zero. By the definition of convergence in norm,

$$\boxed{f_n \to 0 \text{ in } L^1(0,1) \text{ and in } L^2(0,1).}$$

(c) Dominated convergence. For $0 < x < 1$ and $n \geq 1$, we have $0 \leq x/n \leq 1$. Choose $g(x) = 1$ on $(0,1)$.

- Every f_n is continuous, hence measurable.
- We proved that $f_n(x) \to 0$ for every $x \in (0,1)$.
- The bound $|f_n| \leq g$ holds for every n.
- The dominating function is integrable: $\int_0^1 g(x)\,dx = 1$.

The dominated convergence theorem therefore gives

$$\lim_{n\to\infty}\int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx = 0.$$

Key point. Convergence in L^2 is checked using the square root of the integral of the square, not the integral alone.

Practice paper 1 – Exercise 2

Probability and conditioning | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. Let $X, Y \in \{0, 1\}$ have the following joint probability distribution:

	$Y = 0$	$Y = 1$
$X = 0$	$3/8$	$1/8$
$X = 1$	$1/8$	$3/8$

(a) **[2 points]** Find the marginal distributions of X and Y. Are the variables independent? Check one joint probability against the product of its marginals.

(b) **[3 points]** Compute $\mathbb{P}(Y = 1 \mid X = 0)$ and $\mathbb{P}(Y = 1 \mid X = 1)$. Hence find $\mathbb{E}[Y \mid X = 0]$ and $\mathbb{E}[Y \mid X = 1]$.

Hint: a variable taking values 0 and 1 has expectation equal to its probability of being 1.

(c) **[2 points]** Use the tower property to compute $\mathbb{E}[Y]$. Check the result using the marginal distribution of Y.

Practice paper 1 – Solution 2

Detailed worked solution

(a) Marginals and independence. Sum each row to obtain the law of X:

$$\mathbb{P}(X=0)=\frac{3}{8}+\frac{1}{8}=\frac{1}{2}, \qquad \mathbb{P}(X=1)=\frac{1}{8}+\frac{3}{8}=\frac{1}{2}.$$

Sum each column to obtain $\mathbb{P}(Y=0)=\mathbb{P}(Y=1)=1/2$. The four entries are nonnegative and sum to one, as required.

Independence would imply equality between every joint probability and the corresponding product of marginal probabilities. However,

$$\mathbb{P}(X=0,Y=0)=\frac{3}{8}\neq\frac{1}{4}=\mathbb{P}(X=0)\mathbb{P}(Y=0).$$

Therefore X and Y are not independent.

(b) Conditioning. Use $\mathbb{P}(A\mid B)=\mathbb{P}(A\cap B)/\mathbb{P}(B)$:

$$\mathbb{P}(Y=1\mid X=0)=\frac{1/8}{1/2}=\frac{1}{4}, \qquad \mathbb{P}(Y=1\mid X=1)=\frac{3/8}{1/2}=\frac{3}{4}.$$

Since $Y\in\{0,1\}$, its conditional means are

$$\mathbb{E}[Y\mid X=0]=0\cdot\frac{3}{4}+1\cdot\frac{1}{4}=\frac{1}{4},$$

$$\mathbb{E}[Y\mid X=1]=0\cdot\frac{1}{4}+1\cdot\frac{3}{4}=\frac{3}{4}.$$

Equivalently, $\mathbb{E}[Y\mid X]=1/4+X/2$ almost surely.

(c) Tower property. Average the conditional means using the probabilities of the conditioning values:

$$\mathbb{E}[Y]=\mathbb{E}[\mathbb{E}[Y\mid X]]=\frac{1}{4}\cdot\frac{1}{2}+\frac{3}{4}\cdot\frac{1}{2}=\frac{1}{8}+\frac{3}{8}=\boxed{\frac{1}{2}}.$$

Directly from the marginal distribution,

$$\mathbb{E}[Y]=0\mathbb{P}(Y=0)+1\mathbb{P}(Y=1)=\frac{1}{2}.$$

Key point. A conditional probability is obtained by dividing a cell by the total probability of the conditioning row.